ORIGINAL ARTICLE

# A Phase 3 Trial of Semagacestat for Treatment of Alzheimer's Disease

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ABSTRACT

**BACKGROUND**

Alzheimer's disease is characterized by the presence of cortical amyloid-beta (A$\beta$) protein plaques, which result from the sequential action of $\beta$-secretase and $\gamma$-secretase on amyloid precursor protein. Semagacestat is a small-molecule $\gamma$-secretase inhibitor that was developed as a potential treatment for Alzheimer's disease.

**METHODS**

We conducted a double-blind, placebo-controlled trial in which 1537 patients with probable Alzheimer's disease underwent randomization to receive 100 mg of semagacestat, 140 mg of semagacestat, or placebo daily. Changes in cognition from baseline to week 76 were assessed with the use of the cognitive subscale of the Alzheimer's Disease Assessment Scale for cognition (ADAS-cog), on which scores range from 0 to 70 and higher scores indicate greater cognitive impairment, and changes in functioning were assessed with the Alzheimer's Disease Cooperative Study–Activities of Daily Living (ADCS-ADL) scale, on which scores range from 0 to 78 and higher scores indicate better functioning. A mixed-model repeated-measures analysis was used.

**RESULTS**

The trial was terminated before completion on the basis of a recommendation by the data and safety monitoring board. At termination, there were 189 patients in the group receiving placebo, 153 patients in the group receiving 100 mg of semagacestat, and 121 patients in the group receiving 140 mg of semagacestat. The ADAS-cog scores worsened in all three groups (mean change, 6.4 points in the placebo group, 7.5 points in the group receiving 100 mg of the study drug, and 7.8 points in the group receiving 140 mg; P=0.15 and P=0.07, respectively, for the comparison with placebo). The ADCS-ADL scores also worsened in all groups (mean change at week 76, –9.0 points in the placebo group, –10.5 points in the 100-mg group, and –12.6 points in the 140-mg group; P=0.14 and P<0.001, respectively, for the comparison with placebo). Patients treated with semagacestat lost more weight and had more skin cancers and infections, treatment discontinuations due to adverse events, and serious adverse events (P<0.001 for all comparisons with placebo). Laboratory abnormalities included reduced levels of lymphocytes, T cells, immunoglobulins, albumin, total protein, and uric acid and elevated levels of eosinophils, monocytes, and cholesterol; the urine pH was also elevated.

**CONCLUSIONS**

As compared with placebo, semagacestat did not improve cognitive status, and patients receiving the higher dose had significant worsening of functional ability. Semagacestat was associated with more adverse events, including skin cancers and infections. (Funded by Eli Lilly; ClinicalTrials.gov number, NCT00594568.)

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Alzheimer's disease begins decades before the appearance of clinical symptoms, with the deposition of aggregated amyloid-beta (A$\beta$) peptide plaques in the cortex and hippocampus. This protein is cleaved from the amyloid precursor protein (APP) by the sequential action of $\beta$- and $\gamma$-secretases, producing fragments that include A$\beta$1-40 and A$\beta$1-42. Since the accumulation of aggregated A$\beta$ is associated with disease progression, both $\beta$-secretase and $\gamma$-secretase represent potential therapeutic targets. Multiple small molecules can inhibit $\gamma$-secretase in vitro,[1-4] but Notch and other transmembrane proteins are also substrates for $\gamma$-secretase,[1-4] and studies have raised concern that the inhibition of $\gamma$-secretase could interfere with Notch receptor–related nuclear signaling and with the function of cell-surface receptors and proteins involved in embryonic development, hematopoiesis, cell adhesion, and other cell-to-cell contacts. Eli Lilly developed semagacestat as a potential therapy for Alzheimer's disease because studies had shown inhibition of $\gamma$-secretase in vitro, reduction of A$\beta$ synthesis by approximately 25% over a period of 24 hours (at a dose of 100, 140, or 280 mg) in humans (as measured with the use of an A$\beta$ synthesis assay), and reduction of A$\beta$ levels in the brains, cerebrospinal fluid (CSF), and plasma of PDAPP transgenic mice carrying human *APP* with a V717F mutation.[5-8] Semagacestat functionally inhibits $\gamma$-secretase rather than competitively inhibiting the enzyme active sites.[9]

Phase 1 studies in human volunteers without Alzheimer's disease showed a dose-dependent decrease in plasma levels of A$\beta$ after 14 days of treatment with 5 to 50 mg of semagacestat[10]; patients with mild-to-moderate Alzheimer's disease had a significant reduction in plasma levels of A$\beta$ but not in CSF levels of A$\beta$ after 6 weeks of treatment with a maximum daily dose of 40 mg of semagacestat.[7] A small phase 2 safety study testing doses of 100 or 140 mg of semagacestat daily for up to 14 weeks as compared with placebo in patients with mild-to-moderate Alzheimer's disease showed dose-related reductions in plasma levels of A$\beta$40 but no significant reductions in CSF levels (measured after a lumbar puncture performed 6 hours after administration of the last dose).[11] Adverse events affecting the skin and subcutaneous tissue were more common among patients receiving semagacestat than among those receiving placebo ($P=0.05$); semagacestat was also associated with reduced uric acid levels ($P<0.001$), prolongation of the prothrombin time and partial-thromboplastin time ($P<0.05$), and reduced percentages of CD19 cells ($P<0.05$).

On the basis of the possibility that semagacestat could have disease-modifying effects in patients with Alzheimer's disease, Eli Lilly launched a multinational phase 3 trial comparing the two doses used in phase 2 with placebo in a randomized, double-blind trial. This trial was halted by the company after the results of a preplanned interim analysis performed by an independent data and safety monitoring board (DSMB) suggested that semagacestat was associated with worsening of cognition and functioning as well as excess skin cancers and a futility analysis indicated that the trial could not show a benefit with respect to the primary outcomes.

## METHODS

### SCREENING

Otherwise healthy patients 55 years of age or older who had mild-to-moderate Alzheimer's disease without depression were randomly assigned to receive 100 mg or 140 mg of semagacestat or placebo daily for 76 weeks. Depression at screening was assessed with the Geriatric Depression Scale (score range, 0 to 15, with higher scores indicating more severe depression)[12]; a score of 6 or less was considered to indicate the absence of depression. Mild-to-moderate Alzheimer's disease was documented on the basis of a score of 16 to 26 on the Mini–Mental State Examination (MMSE; score range, 0 to 30, with higher scores indicating better cognitive function)[13] and the criteria of the National Institute of Neurological and Communicative Diseases and Stroke–Alzheimer's Disease and Related Disorders Association.[14] Patients being treated with background cholinesterase inhibitors, memantine, or both were admitted to the study.

### OVERSIGHT

The research protocol was approved by the institutional review board at each institution where the trial was conducted, and all participants provided written informed consent. The study protocol and details on informed consent are available with the full text of this article at NEJM.org. All the authors vouch for the completeness and veracity of the data and the data analysis and for



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the fidelity of this report to the protocol. Eli Lilly or its contract research organization collected the data, and the Alzheimer's Disease Cooperative Study (ADCS) performed the analysis and wrote the report, which was approved by the voting members of the Data Analysis and Publication Committee (a collaborative body of the ADCS and Eli Lilly, with the latter not having voting rights) and the ADCS steering committee (for a list of members of both committees, see the Supplementary Appendix, available at NEJM.org).

#### SAFETY ASSESSMENTS

Monitoring for adverse events was performed at each visit. Additional safety assessments included measurement of lymphocytes, examination of the skin, urinalysis, and electrocardiography (ECG). Because semagacestat strongly activates the production of cytochrome P-450 3A (CYP3A), levels of representative drugs likely to be affected by CYP3A were assessed periodically, and donepezil levels were measured at baseline, week 12, and week 52.

#### OUTCOME MEASURES

Efficacy measures included the cognitive subscale of the Alzheimer's Disease Assessment Scale (ADAS-cog; score range, 0 to 70, with higher scores indicating greater cognitive impairment),[15] the Alzheimer's Disease Cooperative Study–Activities of Daily Living (ADCS-ADL) scale (score range, 0 to 78, with higher scores indicating better function),[16] the Clinical Dementia Rating–Sum of Boxes (CDR-SB),[17,18] the Neuropsychiatric Inventory (NPI),[19] the Resource Utilization in Dementia Lite (RUD-Lite) scale,[20] the European Quality of Life–5 Dimensions (EQ-5D) scale (proxy version),[21] and the MMSE.[13]

Apolipoprotein E genotyping was performed at study entry, and lymphocyte phenotyping was performed periodically. Plasma levels of A$\beta$ were assessed throughout the study, and the following studies were performed for a subset of patients at baseline and at week 76 or at the time of early termination: CSF analysis for levels of A$\beta$ and tau protein, volumetric magnetic resonance imaging (MRI), positron-emission tomographic (PET) imaging with the use of $^{18}$F-florbetapir for the visualization of A$\beta$. The analyses of plasma and CSF and the performance of $^{18}$F-fluorodeoxyglucose (FDG) PET and other imaging studies were carried out to obtain evidence of disease modification and pharmacodynamic effect.

#### STATISTICAL ANALYSIS

The primary analysis compared the change in scores on the ADAS-cog and the ADCS-ADL scale from baseline to week 76 in the three study groups. The DSMB charter called for an interim analysis of the coprimary outcomes after 50% of patients had completed 12 months of treatment or had dropped out of the study. DSMB members were initially unaware of group assignments but were to be informed of them if the P value for the differences in coprimary outcomes between any two groups was less than 0.05. When it was determined that the outcomes for the two treatment groups were worse than they were for the control group, the DSMB decided that the futility analysis, originally scheduled to take place after 50% of participants had been in the study for 18 months, should be performed. The DSMB recommended stopping the trial on the basis of the results of the futility analysis.

We used a mixed-model repeated-measures analysis to compare model-adjusted least-squares means at 76 weeks. The model included the fixed effects of baseline scores on the ADAS-cog, ADCS-ADL scale, CDR-SB, MMSE, RUD-Lite scale, NPI, and EQ-5D scale; group assignment (high-dose semagacestat, low-dose semagacestat, or placebo); MMSE score at screening (continuous variable); visit and treatment-by-visit interaction; concomitant treatment with a cholinesterase inhibitor or memantine at baseline (yes or no); and age at baseline. The random effect (random intercept) was "participant." Study visit was treated as a categorical variable. The primary efficacy analysis was based on a modified intention-to-treat population, which included all randomly assigned participants for whom there was at least one postbaseline observation. A secondary analysis was performed for all participants who completed the initial 76-week period of treatment with the study medication.

The coprimary outcomes were the change in scores on the ADAS-cog and the ADCS-ADL scale from baseline to study end point (week 76). Secondary outcomes were the change from baseline in scores on the CDR–SB, MMSE, NPI, RUD-Lite scale, and EQ-5D scale; the values for plasma and CSF levels of A$\beta$ and for CSF levels of tau; MRI volumetric measures; and imaging studies performed with FDG-PET and $^{18}$F-florbetapir–PET.

Safety analyses were based on the full intention-to-treat population, and all biomarker analy-



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ses were calculated with the use of data from patients with at least one postbaseline value. The baseline characteristics of the three study groups were compared with the use of a Fisher–Freeman–Halton test for categorical variables and an analysis of variance for continuous variables. An analysis of covariance was used to assess changes from baseline in plasma and CSF biomarkers and in imaging studies. Safety analyses included summary listings of adverse events, with Fisher's exact test used for pairwise comparisons. R Statistical software, version 2.14.1 (R Development Core Team 2011), was used for all statistical analyses. For the coprimary outcomes, the analysis was duplicated with the use of SAS software, version 9 (SAS Institute). The significance level was set at P<0.05. All statistical testing was two-sided. Table S1 in the Supplementary Appendix shows the slight differences between the ADCS analysis plan and the original Eli Lilly analysis plan.

## RESULTS

### COMPLETION RATES AND BASELINE CHARACTERISTICS

A total of 2009 patients were screened for study participation and 1537 underwent randomization (Fig. 1). The most common reason for not proceeding to randomization was failure to meet the entry criteria (accounting for 77% of screened patients who did not meet the entry criteria). Completion rates were 38% for participants in the placebo group, 30% for those receiving 100 mg of semagacestat, and 23% for those receiving 140 mg of semagacestat; most discontinuations were due to the termination of the trial. Discontinuation due to adverse events was higher in the groups receiving semagacestat (P<0.001).

There were no significant differences among the groups with regard to baseline characteristics (Table 1). Approximately 82% of participants were being treated with a cholinesterase inhibitor, 33% with memantine, and 29% with both.

### COGNITIVE AND CLINICAL OUTCOMES

*Coprimary Outcomes*

Table 2 shows modeled changes from baseline for the coprimary and secondary outcomes at 76 weeks in the modified intention-to-treat population (also see Fig. S2 through S7 in the Supplementary Appendix). Cognitive status worsened in all three groups (mean change in the ADAS-cog score at week 76, 6.4 points in the placebo group, 7.5 points in the group receiving 100 mg of semagacestat, and 7.8 points in the group receiving 140 mg of semagacestat; P=0.15 for the comparison of 100 mg with placebo, and P=0.07 for the comparison of 140 mg with placebo). Functional status also worsened in all three groups (mean change in the ADCS-ADL score at week 76, –9.0 points with placebo, –10.5 points with 100 mg of semagacestat, and –12.6 points with 140 mg of semagacestat; P=0.14 for the comparison of 100 mg with placebo, and P<0.001 for the comparison of 140 mg with placebo). An analysis of the 679 patients who completed 76 weeks of treatment showed no significant differences in scores on either the ADAS-cog or the ADCS-ADL scale among the three groups.

*Secondary Outcomes*

Scores on the CDR-SB showed worsening severity of dementia in all three study groups, with greater deterioration in the two semagacestat groups than in the placebo group (Fig. S3 in the Supplementary Appendix). MMSE and NPI scores showed significant worsening in the group receiving 140 mg of semagacestat as compared with the placebo group in the modified intention-to-treat population (Fig. S4 and S5 in the Supplementary Appendix). These differences were not observed in an analysis restricted to participants who completed the 76-week period of treatment. There were no significant differences among groups with respect to the RUD-Lite score (Table S7 in the Supplementary Appendix). As compared with the placebo group, the group receiving 140 mg of semagacestat had greater worsening on the EQ-5D scale (U.S. and U.K. index scores) at weeks 52 and 76, and the group receiving 100 mg of semagacestat had greater worsening at weeks 28, 52, and 76 (Table S7 in the Supplementary Appendix).

### SAFETY AND SIDE-EFFECT PROFILE

*Weight*

By week 52, patients in both active-treatment groups had lost weight (mean [±SD] change, –1.6±4.4 kg in the group receiving 100 mg of semagacestat and –1.6±4.7 kg in the group receiving 140 mg), whereas those in the placebo group had gained an average of 0.4±3.9 kg by week 52 (P<0.001 for both comparisons). The percentage



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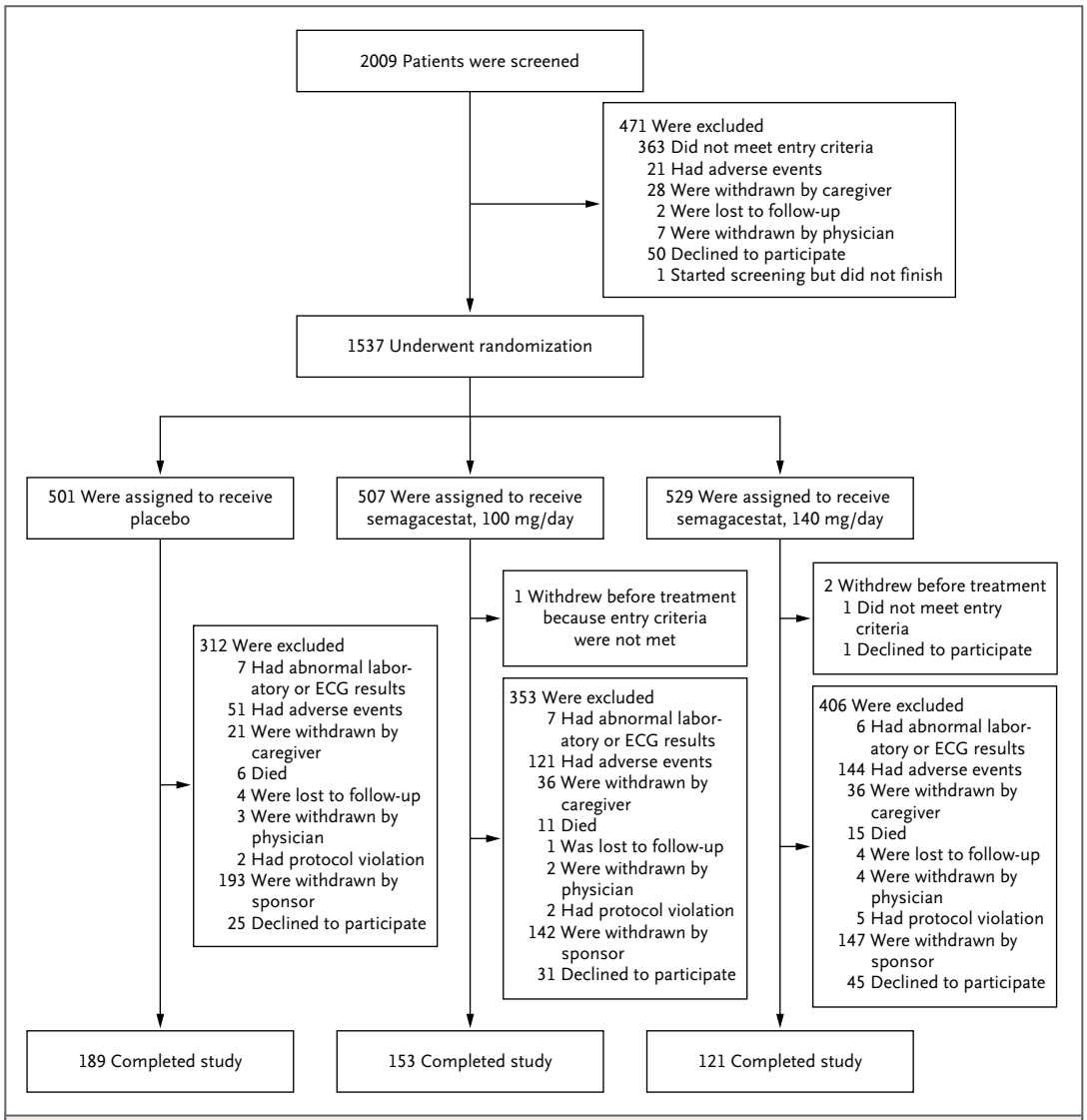


**Figure 1. Screening, Randomization, and Follow-up during the Initial 76-Week Treatment Period.**
Patients were withdrawn by the study sponsor because of adverse effects of the study drug and futility (according to the futility analysis) reported by the data and safety monitoring board. ECG denotes electrocardiographic.

of patients who lost more than 7% of their baseline weight by week 52 was 7% in the placebo group, as compared with 19% in the lower-dose semagacestat group and 20% in the higher-dose semagacestat group (P<0.001 for both comparisons). The percentage of patients who lost more than 4% of their baseline weight at week 52 was 15% in the placebo group, as compared with 31% in the lower-dose semagacestat group and 33% in the higher-dose semagacestat group (P<0.001 for both comparisons).[22]

*ECG Results*

ECGs showed a small increase from baseline in the QT interval corrected for heart rate at week 72 in the two treatment groups (mean increase, 3.5±15.3 msec in the lower-dose semagacestat group and 3.7±15.4 msec in the higher-dose semagacestat group, vs. a decrease of 0.44±13.7 msec in the placebo group). The increase was significantly larger in the two active-treatment groups than in the placebo group (P<0.001 for both comparisons).



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**Table 1. Baseline Characteristics of the Study Participants.***

| Variable | Placebo (N=501) | Semagacestat, 100 mg (N=506) | Semagacestat, 140 mg (N=527) | Total (N=1534) |
|---|---|---|---|---|
| Age — yr | 73.3±8.1 | 73.0±8.0 | 73.3±8.5 | 73.2±8.2 |
| Male sex — no. (%) | 223 (45) | 227 (45) | 264 (50) | 714 (47) |
| Medication — no. (%) | | | | |
| Cholinesterase inhibitor use only | 274 (55) | 263 (52) | 272 (52) | 809 (53) |
| Memantine use only | 21 (4) | 21 (4) | 16 (3) | 58 (4) |
| Cholinesterase inhibitor and memantine use | 139 (28) | 149 (30) | 101 (31) | 449 (29) |
| MMSE score† | 20.9±3.6 | 20.8±3.5 | 20.7±3.5 | 20.8±3.5 |
| ADAS-cog score‡ | 22.8±9.1 | 22.5±8.9 | 23.1±8.8 | 22.9±9.0 |
| *APOE* ε4 carrier — no./total. no. (%) | 256/441 (58) | 291/459 (63) | 268/463 (58) | 815/1363 (60) |

* Plus–minus values are means ±SD. There were no significant differences among groups with regard to baseline characteristics.

† In the Mini–Mental State Examination (MMSE), the range of scores is 0 to 30, and higher scores indicate better cognitive function.

‡ On the cognitive subscale of the Alzheimer's Disease Assessment Scale (ADAS-cog), the range of scores is 0 to 70, and higher scores indicate greater cognitive impairment.

*Adverse Events and Deaths*

Adverse events were more common in the two semagacestat groups than in the placebo group (P<0.001 for each dose vs. placebo). There were more cancers and more skin and subcutaneous-tissue disorders in the semagacestat groups (Table 3). The following adverse events, classified according to the preferred terms from the *Medical Dictionary for Regulatory Activities* (MedDRA), version 13.1, occurred in at least 2% of patients in any group and at a rate that was at least twice as high in the combined active-treatment groups as in the placebo group: skin cancers (basal cell and squamous cell), gastrointestinal symptoms (decreased appetite, nausea, vomiting, and weight loss), hair and eyelash hypopigmentation, epistaxis, rashes and pruritus, alopecia, and syncope (Table 3). The percentage of patients who discontinued the study drug because of adverse events was higher with semagacestat than with placebo (26% with 100 mg and 30% with 140 mg vs. 11% with placebo, P<0.001 for both comparisons). There were more cancers (especially nonmelanoma skin cancer) and more infections in the two groups receiving semagacestat (Table S4 in the Supplementary Appendix).

Several laboratory findings pointed to drug-related and clinically significant effects of semagacestat (Table S5 in the Supplementary Appendix). There was indirect evidence of Fanconi's syndrome with a dose-related reduction in serum uric acid and phosphorus levels (presumably because of increased renal excretion) and impaired urinary acidification. Patients were not assessed for glycosuria.

There was evidence that semagacestat was associated with hepatocellular injury, with increased serum alanine aminotransferase levels and with smaller elevations in levels of aspartate aminotransferase. Levels of bilirubin (primarily direct bilirubin) dropped in both groups receiving semagacestat, and there were dose-related increases in cholesterol levels. Immune function was also altered. The most notable change was a reduction in IgG levels that was not related to the dose; in addition, there was a transient elevation in IgM levels that was unrelated to the dose and a transient dose-related reduction in IgA levels. Lymphocyte counts, particularly CD19 cells, declined (a change that was not related to the dose), and there were small, dose-dependent increases in monocyte and eosinophil counts.

The rate of serious adverse events was higher in the semagacestat groups than in the placebo group: 24% in the group receiving 100 mg and 25% in the group receiving 140 mg, as compared with 14% in the group receiving placebo (P<0.001 for both comparisons) (Table S6 in the Supplementary Appendix). More deaths occurred among patients receiving semagacestat (9 in the group receiving 100 mg and 14 in the group receiving 140 mg vs. 6 in the group receiving placebo, P=0.25 for both comparisons) (Table S8 in the Supplementary Appendix).



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**Table 2. Estimated Mean Change from Baseline for the Coprimary and Secondary Outcomes, According to a Mixed-Model Repeated-Measures Analysis.***

| Outcome | Placebo | Semagacestat, 100 mg | Semagacestat, 140 mg | P Values | |
|---|---|---|---|---|---|
| | | | | Semagacestat, 100 mg, vs. Placebo | Semagacestat, 140 mg, vs. Placebo |
| ADAS-cog score | | | | 0.15 | 0.07 |
| No. of participants with results | 486 | 483 | 497 | | |
| Mean change in score (95% CI) | 6.4 (5.48 to 7.40) | 7.5 (6.44 to 8.53) | 7.8 (6.72 to 8.85) | | |
| ADCS–ADL† | | | | 0.14 | <0.001 |
| No. of participants with results | 480 | 481 | 490 | | |
| Mean change in score (95% CI) | −9.0 (−10.37 to −7.67) | −10.5 (−11.94 to −9.07) | −12.6 (−14.1 to −11.2) | | |
| CDR-SB‡ | | | | 0.06 | <0.01 |
| No. of participants with results | 485 | 480 | 494 | | |
| Mean change in score (95% CI) | 2.4 (2.06 to 2.67) | 2.8 (2.47 to 3.13) | 3.1 (2.73 to 3.41) | | |
| NPI§ | | | | 0.28 | 0.05 |
| No. of participants with results | 473 | 463 | 472 | | |
| Mean change in score (95% CI) | 1.9 (0.69 to 3.12) | 2.9 (1.58 to 4.21) | 3.7 (2.36 to 5.08) | | |
| MMSE | | | | 0.23 | 0.03 |
| No. of participants with results | 400 | 328 | 303 | | |
| Mean change in score (95% CI) | −3.4 (−3.95 to −2.86) | −3.9 (−4.51 to −3.30) | −4.3 (−4.99 to −3.68) | | |

* In this mixed-model repeated-measures analysis, the dependent variable was the change from the baseline score at each postbaseline visit during the initial treatment period. Fixed effects included the baseline score, treatment, MMSE score at screening, visit, treatment effect according to visit, concomitant use of an acetylcholinesterase inhibitor (usually donepezil) or memantine at baseline (yes or no), and age at baseline. Study visit was treated as a categorical variable. "Participant" was included in the model as the random effect. An unstructured covariance matrix was used to model the within-participant variance–covariance errors. The primary comparisons of interest were those of 140 mg of semagacestat versus placebo, 100 mg of semagacestat versus placebo, and 140 mg versus 100 mg of semagacestat at week 76. The numbers of participants are based on those who had a baseline score and at least one postbaseline score for each instrument. CI denotes confidence interval.

† For the Alzheimer's Disease Cooperative Study–Activities of Daily Living (ADCS-ADL) scale, the range is 0 to 78, and higher scores indicate better function.

‡ For the Clinical Dementia Rating–Sum of Boxes (CDR-SB), the range is 0 to 18, and higher scores indicate poorer function.

§ For the Neuropsychiatric Inventory (NPI), the range is 0 to 144, and higher scores indicate poorer function.

*Donepezil Levels*

Plasma donepezil levels decreased by approximately 30% between baseline and week 52 in both semagacestat groups (−20.3±19.2 ng per milliliter in the lower-dose group and −20.4±19.7 ng per milliliter in the higher-dose group, vs. 0.46±18.4 ng per milliliter in the placebo group; P<0.001 for both comparisons). Reductions in donepezil levels did not correlate with worsening of the coprimary or secondary outcomes.

*Biologic Markers and Neuroimaging*

Among the 1467 participants in whom plasma levels of Aβ were measured at baseline, 844 had levels measured at week 76 or at early termination. In the group receiving 100 mg of semagacestat, there were reductions from baseline in levels of Aβ40 (mean, 38%; median, 47%) and Aβ42 (mean, 5%; median, 9%); in the group receiving 140 mg, there were greater reductions in levels of both Aβ40 (mean, 48%; median, 54%) and Aβ42 (mean, 18%; median, 24%) (P<0.001 by analysis of covariance for all comparisons). There was no significant change from baseline in levels of Aβ40 or Aβ42 in the placebo group.

Levels of Aβ in the CSF were available for 110 participants at baseline and for 47 of these participants at week 76 or early termination. CSF levels of Aβ40 and Aβ42 (measured 24 to 36 hours after the administration of the study drug or placebo) did not change significantly in any group. Levels of tau protein in the CSF were available for 109 participants at baseline and for 47 of these participants at week 76 or early termination. There were no significant changes from baseline in any of the three study groups.



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**Table 3. Summary of Adverse Events Occurring during the Study Treatment, According to the MedDRA System or Organ Class and Preferred Term.***

| Event | Placebo (N=501) | Semagacestat | | | Total (N=1534) |
|---|---|---|---|---|---|
| | | 100 mg (N=506) | 140 mg (N=527) | Combined (N=1033) | |
| | | *percent of participants* | | | |
| System or organ class | | | | | |
| Neoplasms — benign, malignant, or unspecified | 5 | 15 | 16 | 15 | 12 |
| Skin or subcutaneous-tissue disorders | 21 | 45 | 52 | 48 | 39 |
| Preferred term | | | | | |
| Alopecia | 0 | 1 | 5 | 3 | 2 |
| Basal-cell carcinoma | 1 | 3 | 5 | 4 | 3 |
| Decreased appetite | 3 | 7 | 11 | 9 | 7 |
| Epistaxis | 1 | 3 | 3 | 3 | 2 |
| Eyelash discoloration | 0 | 2 | 5 | 3 | 2 |
| Hair-color changes | 1 | 13 | 19 | 16 | 11 |
| Nausea | 5 | 11 | 12 | 11 | 9 |
| Pruritus | 2 | 3 | 4 | 3 | 3 |
| Rash | | | | | |
| Erythematous | 2 | 5 | 5 | 5 | 4 |
| Macular | 4 | 7 | 8 | 8 | 6 |
| Maculopapular | 1 | 3 | 5 | 4 | 3 |
| Papular | 1 | 3 | 4 | 4 | 3 |
| Skin lesion | 1 | 3 | 3 | 3 | 2 |
| Squamous-cell carcinoma of skin | 1 | 6 | 5 | 5 | 4 |
| Syncope | 1 | 3 | 3 | 3 | 3 |
| Vomiting | 4 | 10 | 9 | 9 | 7 |
| Weight decrease | 3 | 5 | 9 | 7 | 6 |

* The listed adverse events are those with an incidence of at least 2% in any one group and a rate in the combined semagacestat groups that was at least two times as high as the rate in the placebo group. MedDRA denotes *Medical Dictionary for Regulatory Activities*, version 13.1.

Levels of phospho-tau in the CSF increased by 16% in the placebo group and declined in the groups receiving semagacestat, with declines of 8% in the lower-dose group and 4% in the higher-dose group ($P<0.001$ for both comparisons with placebo).

Among the 208 participants for whom volumetric MRI scans were obtained at baseline, the availability of end-point data differed according to imaging outcome, from a low of 120 scans available for entorhinal cortex volume to 208 scans available for hippocampal volume. A few scans showed greater volumetric loss in patients receiving 140 mg of semagacestat than in those receiving placebo, but the results were inconsistent across imaging regions, and baseline measures showed significant imbalances across treatment groups. One patient had vasogenic edema at baseline, but new vasogenic edema was not seen in patients in any of the groups. PET scans were available for 108 participants at baseline and for 59 of these participants at week 76 or early termination. There were no significant differences among the three groups in changes from baseline FDG-PET or $^{18}$F-florbetapir–PET measurements.

*Reversal of Adverse Events and Clinical Worsening*

After the termination of drug dosing, data were available for 58% of participants (290 of 501)



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who were randomly assigned to placebo, 44% (223 of 506) who were assigned to 100 mg of semagacestat, and 41% (214 of 527) who were assigned to 140 mg of semagacestat (including participants who completed the initial 76-week period of study treatment and those who were still receiving the study drug when the trial was stopped). At the end of the 32-week safety-extension phase, changes from baseline in ADAS-cog and ADCS-ADL scores were not significantly different across the three groups. There were no longer significant changes from baseline in body weight in the two semagacestat groups. Some (14.5%) of the adverse events that were present at the end of treatment were reduced in severity or reversed by the end of the extension phase. However, the changes from baseline in levels of phosphorus, IgG, and IgA and in white-cell counts were significant, and alkaline phosphatase levels were elevated in the two semagacestat groups.

## DISCUSSION

This randomized, double-blind, placebo-controlled trial showed no benefit of the $\gamma$-secretase inhibitor semagacestat for the treatment of mild-to-moderate Alzheimer's disease. Indeed, treatment with semagacestat was associated with dose-related clinical worsening on multiple primary and secondary outcome measures, including measures of activities of daily living measure (ADCS-ADL score), global functioning (CDR-SB), secondary cognitive functioning (MMSE), and quality of life (EQ-5D) in the group receiving the higher dose of semagacestat. Although donepezil levels were reduced in both groups receiving semagacestat (levels were probably elevated as a result of CYP3A4 induction), this pharmacologic effect does not explain the clinical worsening.

The study was stopped early after a futility analysis, which was conducted earlier than planned, once clinical worsening and safety concerns were identified. Patients receiving semagacestat had more adverse events and more serious adverse events and were more likely to discontinue the study because of adverse events than patients receiving placebo. There were also more gastrointestinal symptoms, infections, and nonmelanoma skin cancers in the semagacestat groups. Changes from baseline laboratory values that were associated with the drug included reductions in IgG and IgA levels and alterations of immune-cell populations (reduced levels of CD19 lymphocytes, reduced levels of monocytes, and eosinophilia). Findings consistent with a drug-induced Fanconi's syndrome were also detected. At the end of the 32-week extension phase of the study, many of the renal and immunologic alterations associated with treatment had not fully resolved.

These disappointing results call for careful further analysis of the data in order to learn as much as possible about the reasons for the clinical worsening. Since Notch and other proteins are also substrates for $\gamma$-secretase,[1-4] the inhibition of $\gamma$-secretase can interfere with Notch receptor–related nuclear signaling and with the function of cell-surface receptors and the proteins involved in embryonic development, hematopoiesis, cell adhesion, and other cell–cell contacts. A recent report suggests that most, if not all, such inhibitors also inhibit Notch, and that semagacestat may be more selective for Notch than for APP processing by $\gamma$-secretase.[23] Since the adverse-event profile of the drug included alterations of immune cells, gastrointestinal symptoms, skin reactions, and skin cancers, it is plausible that the inhibition of Notch accounted for many of these adverse events. The question that is more difficult to answer is whether interference with Notch, or some other receptor or protein in the brain, accounted for the clinical worsening seen in the groups receiving semagacestat.

Alternatively, the clinical worsening may have been related to the effects of semagacestat on its target — APP metabolism by the $\gamma$-secretase complex. There is evidence from the data on biologic markers that this target was engaged — specifically, there were reductions in plasma levels of A$\beta$40 and A$\beta$42. However, the lack of significant reductions in CSF levels of these A$\beta$ proteins at the time point selected for assessment suggests that the target was engaged peripherally rather than in the brain.[24] The absence of treatment-related changes detected in other brain imaging studies, such as FDG-PET, amyloid imaging with PET, and volumetric MRI studies, could be due to a lack of relevant effects, an insufficient magnitude of effects, imbalances across the subgroups at baseline, or an inadequate sample size or assessment period. However, the absence of between-group differences in these outcomes also suggests that amyloid reduction by semagacestat did not have a directly detrimental effect on the brain.



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A notable aspect of the current report is the willingness of the sponsor to provide its proprietary data sets to an independent committee for analysis and publication. An important limitation of the study is that it was halted early, which meant that the number of participants and the duration of their participation were limited. Another, concurrent study of the effect of semagacestat in patients with mild-to-moderate Alzheimer's disease (140 mg vs. placebo) was also terminated early because of the findings reported here.

In conclusion, the $\gamma$-secretase inhibitor semagacestat was associated with significant toxic effects that were probably related to its effects on proteins other than APP. The drug was also associated with clinical worsening. The question of whether this worsening represents an on-target effect of central $\gamma$-secretase inhibition and central A$\beta$ reduction or an unintended effect related to the known engagement of other inhibitory substrates remains unanswered.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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